COURSE TITLE : **PARALLEL PROCESSING**
COURSE CODE : **6152**
COURSE CATEGORY : **E**
PERIODS/WEEK : **4**
PERIODS/SEMESTER : **60**
CREDITS : **4**

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Parallel Computing	15
2	Parallel Algorithm Design	15
3	Analytical Modeling of Parallel Programs	15
4	Programming Shared Address Space Platforms	15

Course General Outcomes:

Sl.	G.O	On completion of this course the student will be able :
1	1	To Understand Parallel Processing
2	1	To Understand Parallel Algorithm Design
3	1	To Understand Parallel Program Design
4	1	To Understand Programming in Shared Address Space Platforms

Specific Outcomes:

MODULE – I Parallel Computing

1.1 To Understand Parallel Processing

1.1.1 Introduction to Parallel Computing
1.1.2 Scope of Parallel Computing
1.1.3 Parallel Programming Platforms –
1.1.4 Limitations of Memory System Performance
1.1.5 Physical Organization of Parallel Platforms
1.1.6 Parallel Network Topologies

MODULE – II Parallel Algorithm Design

2.1 To Understand Parallel Algorithm Design

2.1.1 Discuss Decomposition, Tasks, and Dependency Graphs
2.1.2 Discuss Granularity, Concurrency, and Task-Interaction
2.1.31 Describe Processes and Mapping
2.1.4 Compare Processes versus Processors
2.1.5 Discuss different Decomposition Techniques
2.1.6 Characteristics of Tasks and Interactions
2.1.7 Methods for Containing Interaction Overheads
2.1..8 Discuss different Parallel Algorithm Models

MODULE – III Analytical Modeling of Parallel Programs

3.1 **To Understand Analytical Modeling of Parallel Programs**

3.1.1 Sources of Overhead in Parallel Programs
3.1.2 Discuss Performance Metrics for Parallel Systems
3.1.3 Describe the Effect of Granularity on Performance
3.1.4 Scalability of Parallel Systems
3.1.5 Programming Using the Message-Passing Paradigm
3.1.6 Principles of Message-Passing Programming
3.1.7 Discuss the Message Passing Interface
3.1.8 Describe Sending and Receiving Messages

MODULE – IV Programming Shared Address Space Platforms

4.1 Programming Shared Address Space Platforms

4.1.1 Programming Shared Address Space Platforms
4.1.2 The POSIX Thread API
4.1.3 Thread Basics: Creation and Termination
4.1.4 Synchronization Primitives in Pthreads
4.1.5 Controlling Thread and Synchronization Attributes
4.1.6 Thread Cancellation
4.1.7 Composite Synchronization Constructs

4.1.8 Tips for Designing Asynchronous Programs
4.1.9 OpenMP: a Standard for Directive Based Parallel Programming

CONTENT DETAILS

MODULE – I Parallel Computing

Introduction to Parallel Computing Motivating Parallelism - Scope of Parallel Computing - Parallel Programming Platforms - Implicit Parallelism: Trends in Microprocessor Architectures - Limitations of Memory System Performance - Physical Organization of Parallel Platforms – network topologies

MODULE – II Parallel Algorithm Design

Principles of Parallel Algorithm Design Preliminaries - Decomposition Techniques- Recursive Decomposition, Data Decomposition ,Exploratory Decomposition, Speculative Decomposition , Hybrid Decompositions – Characteristics of Tasks and Interactions - Methods for Containing Interaction Overheads -Parallel Algorithm Models - Data-Parallel Model, Task Graph Model, Work Pool Model , Master-Slave Model, Pipeline or Producer-Consumer Model, Hybrid Models

MODULE – III Analytical Modeling of Parallel Programs

Analytical Modeling of Parallel Programs Sources of Overhead in Parallel Programs - Performance Metrics for Parallel Systems - The Effect of Granularity on Performance -Scalability of Parallel Systems - Minimum Execution Time and Minimum Cost-Optimal Execution Time - Asymptotic Analysis of Parallel Programs - Other Scalability Metrics – Principles of Message-Passing Programming - The Building Blocks: Send and Receive Operations - MPI: the Message Passing Interface

MODULE – IV Programming Shared Address Space Platforms

Programming Shared Address Space Platforms Thread Basics - The POSIX Thread API - Thread Basics: Creation and Termination - Synchronization Primitives in Pthreads -Controlling Thread and Synchronization Attributes - Thread Cancellation - Composite Synchronization Constructs - Tips for Designing Asynchronous Programs - OpenMP: a Standard for Directive Based Parallel Programming

Text Book(s)

1. : Introduction to Parallel Computing (2nd Edition) Ananth Grama (Author), George Karypis (Author), Vipin Kumar (Author), Anshul Gupta (Author)

Reference:

1. An Introduction to Parallel Programming Peter Pacheco (Author)